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RISK EVALUATION AND RESPONSE PROCEDURE

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1.0 PURPOSE

The purpose of this document is to define the QatarEnergy-IS processes by which potential consequences and probabilities of identified Event Scenarios and associated risks in QatarEnergy-IS activities are assessed and managed.

The Procedure identifies requirements for the establishment and maintenance of an QatarEnergy-IS Risk Inventory, in accordance with the requirements of the QatarEnergy-IS HSE Management System (HSEMS), and the Risk Management Standard.

2.0 SCOPE AND APPLICABILITY

This Procedure applies to Employees, contractors, and subcontractors, whether engaged directly by QatarEnergy or by others, who will perform work or services of any kind at any QatarEnergy-IS facilities and/or locations.

The use of this Procedure is mandatory for HSE Risk Evaluation and Response activities carried out in QatarEnergy-IS.

3.0 ROLES AND RESPONSIBILITIES

3.1 PHR TEAM

The members of a PHR Team shall:

- a. Apply the appropriate technique (e.g. What-if Risk Assessment, HAZOP).
- b. Identify Hazards.
- c. Develop credible Event Scenarios for the identified hazards.
- d. Assign unmitigated Consequence Levels, based on the Risk Matrix, for each Event Scenario. Assign Probability Level for each credible consequence category. Determine intrinsic controls and recommendations for Event Scenarios and use the Risk Matrix to determine Initial Risk Levels.
- e. Evaluate PEC for all Event Scenarios that have a level 5 consequence assigned.
- f. Determine any possible Additional Controls and Recommendations and reevaluate, utilizing the Risk Matrix, to determine the Residual Risk Level and confirm 'ALARP'.

3.2 TECHNICAL HSE ENGINEER

The Technical HSE Engineer shall:

- a. Prepare a HSE Risk Register for all 'C-with/PEC', 'D' and 'E' Level HSE Risks verified by the Risk Verifier.
- b. Support the QatarEnergy-IS functions in regard to satisfactory implementation of the RM programs.

3.3 TECHNICAL HSE ADMINISTRATOR

The Technical HSE Administrator shall:

- a. Be the scribe at PHR reviews.

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- b. Obtain required review and approvals and issue PHR reports on reviews attended.
- c. Maintain a register of all PHR reviews carried out, distribute actions to relevant parties, and issue a monthly management report summary.
- d. Attach PHR reports into relevant SAP MOC record.
- e. Generally administer the RM / PHR program in QatarEnergy-IS.

3.4 TECHNICAL HSE TEAM

The Technical HSE Team are designated members of the HSE Department for:

- a. The review of PHR reports for Clarity, Consistency and Completeness.
- b. Issue out reports to all concerned Parties for their acknowledgement, approval and action close-out.
- c. Verification of PHRs with Risk Levels of 'C-with/PEC', 'D' and/or E
- d. Update QatarEnergy-IS HSE Risk Register(s) and related documents.

3.5 OFFSHORE INSTALLATION SUPERINTENDENT (OIS)/ OPERATIONAL FOCAL POINT

The Offshore Installation Superintendent/ Operational Focal Point shall:

- a. Receive all 'A', 'B', 'C', 'D' and 'E' Risk Levels for awareness.
- b. For What-if risk assessments: receive all Risk assessment report for sign off.
- c. Follow-up on the close-out of identified actions.

3.6 ASSET MANAGER (OI)

The Asset Manager shall:

- a. For What-if risk assessments: Receive all 'C-with/PEC', 'D' and 'E' Risk Levels at Rev. B for acknowledgement and approval.
- b. Receive notifications for 'C-with/PEC', 'D' and 'E' Risk Levels from all other PHA studies following risk verification exercises and sign-off/ approve.
- c. Receive update of the QatarEnergy-IS HSE Risk Register(s) as required.

3.7 HSE MANAGER (OIH)

The HSE Manager shall:

- a. For What-if risk assessments: Receive all 'C-with/PEC', 'D' and 'E' Risk Levels at Rev. B for acknowledgement and approval.
- b. Receive notifications for 'C-with/PEC', 'D' and 'E' Risk Levels from all other PHA studies following risk verification exercises and sign-off/ approve.
- c. Receive update of the QatarEnergy-IS HSE Risk Register(s) as required.

3.8 FIELD OPERATIONS MANAGER (OIF)

The Field Operations Manager shall:

- a. For What-if risk assessments: Receive all 'C-with/PEC', 'D' and 'E' Risk Levels at Rev. B for acknowledgement and approval.

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- b. Receive notifications for 'C-with/PEC', 'D' and 'E' Risk Levels from all other PHA studies following risk verification exercises and sign-off/ approve.
- c. Receive update of the QatarEnergy-IS HSE Risk Register(s) as required.

4.0 PROCEDURE

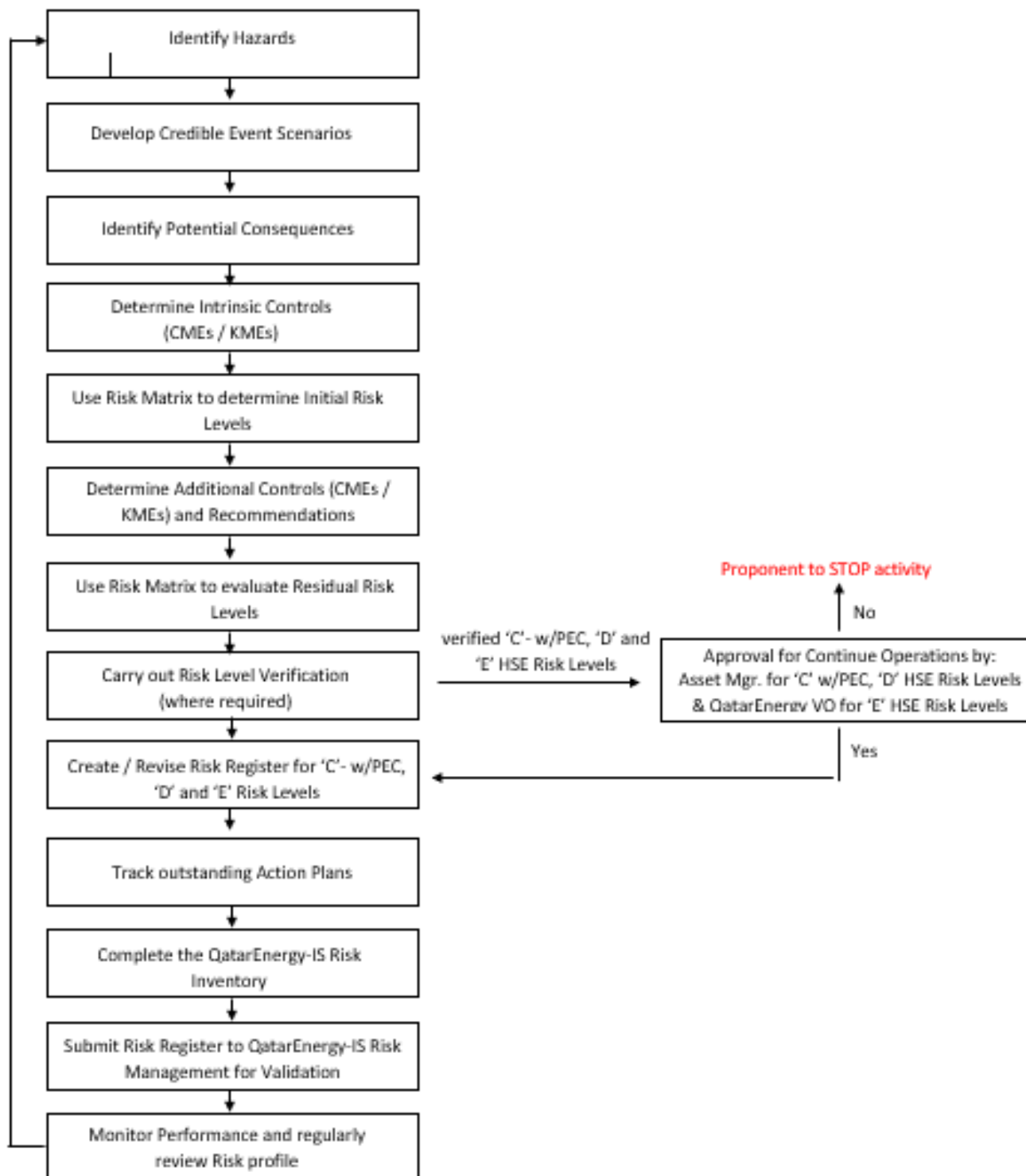
QatarEnergy-IS, in line with the HSE Management System and Risk Management Standard, requires HSE Risks to be identified, evaluated, reported and appropriately managed. Credible Event Scenarios have been, and will continue to be, established, and their resulting consequences and probabilities recorded, applying the QatarEnergy-IS HSE Risk Matrix to determine Risk Levels.

The Risk Response specified for each Risk Level shall be in accordance with QatarEnergy-IS HSE Risk Matrix, Appendix 1, henceforth termed the 'Risk Matrix'.

The QatarEnergy-IS Risk Assessment Cycle is shown in Figure 1 below:

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FIGURE 1. QATARENERGY-IS RISK ASSESSMENT CYCLE



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4.1 HSE RISK MANAGEMENT PROGRAM ELEMENTS

4.1.1 Hazard Identification¹

Hazards are situations that pose a level of threat to personnel, environment, property, or revenue. These represent Risks that may be identified through focused hazard identification activities such as Process Hazard Reviews (PHRs) or other activities for example mechanical integrity inspections, facility walk-through inspections, Incident investigations, internal and external Audits, property insurance surveys, and Lessons Learned communications.

The Risk identification and Risk evaluation activities shall be combined when using detailed process hazard reviews such as What-If and Hazard and Operability Study (HAZOP).

4.1.2 Event Scenarios²

An Event Scenario consists of credible initiating/escalating event(s) that can lead to an estimated consequence of exposure of that hazard being realized. For each identified hazard, an Event Scenario shall be developed if there are any of the following credible consequences:

- Human injury / illness at the facility.
- Environmental Impairment.
- Facility property damage and loss of revenue (excluding clean-up/remediation).
- Impacts outside the facility.

Where there is no credible potential for exposure within the above consequences, there is no credible Event Scenario.

A Process Event Scenario typically consists of an initiating event (e.g. inadvertent valve closure resulting in overpressure and a subsequent series of events leading to 'Loss of Containment'). A release scenario shall be developed, and then an additional series of events or situations shall be identified that could result in each of the consequences as identified in the Risk Matrix (ref. Appendix C).

4.1.3 Consequence Levels

Each credible Consequence Level of 1 to 5, based on the Risk Matrix, shall be assigned for each Event Scenario, Appendix C Risk Matrix. Assignment of initial Consequence Levels shall not take into account mitigation measures or elements. Mitigation measures / elements shall only be taken into account when assigning Probability Levels – see section 4.1.4 below.

However, if the Event Scenario is credible but not “worst case” (e.g. if people are seldom in the immediate area impacted by a release, say a wellhead on a Remote Jacket), the PHR Team should not automatically assume an injury consequence.

When a PHR team determines that a combined Property Damage and Loss of Revenue (PD/LOR) consequence is potentially a Level 3 or higher, the team shall estimate the PD/LOR.

¹ See CORP-HSE-PRC-009, Corporate Procedure For Hazard Identification Study

² See IS-HSE-GDL-310, Developing Process Event Scenarios Guideline

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4.1.4 Probability Levels

Probability Levels represent the presence of measures / elements that influence the potential for an Event Scenario occurring.

Typically, in a process facility there is a probability that containment will be lost, and a subsequent lower probability of each consequence occurring (e.g., for an explosion scenario that impacts people, the release must occur, the release must be ignited and people must occupy the overpressure area).

Each Event Scenario shall be assigned a Probability Level for each credible consequence category, in accordance with the Risk Matrix (Appendix C).

For Operational Risks, Critical Mitigating Elements (CME³), shall be identified, unless CMEs are not appropriate.

Quantitative Risk Analysis (QRA) shall be used in case the CME approach cannot be applied.

Failing either of the above CME or QRA approaches, then the “historical approach” to assigning Probability Level will be applied.

4.1.5 Initial, Residual Risk Levels & Action items

The initial Risk Level of an Event Scenario shall be derived from the assigned Consequence Level and an assumed Probability Level 5. The Probability Level shall then be reduced by one level per identified robust Intrinsic Control, which determines the Risk Level in the Risk Matrix of each potential HSE consequence of the Event Scenario.

The PHR Team shall review details of the scenario and identify any Additional Controls that they consider should be adopted in order to reduce the initial Risk level to that which they consider ‘ALARP’ for the scenario. This shall then be followed by a re-evaluation of the assigned Risk Level, using the Risk Matrix, to give a ‘Residual Risk’ Level. The PHR Team shall also identify any Recommendations that they consider appropriate.

Action items in QatarEnergy-IS are of two types:

- **Additional Controls:** Facility or change-specific controls, (CME), usually identified in the PHR reviews in order to reduce risk, from the levels achieved by the Intrinsic controls, or to support an existing, Intrinsic or ‘Additional control’ (KME). ‘Additional Controls’ should be further categorized to eliminate, prevent, detect, mitigate, respond, and recover with regard to the Event Scenario.
- ‘Additional Controls’ **shall be defined in clear, explicit terms**; use of vague language and conditional terms e.g. ‘Consider’; shall be avoided. An Additional Control should reflect a reduction in risk and give the responsibility to someone to take action. Well-defined and clear action item wording and requirements are essential. In all PHRs any CME / KME (Additional Control), Equipment that is identified by the PHR Team to be HSE Critical shall be recorded as ‘HSE Critical Equipment’ in the PHR report.

³ See IS-HSE-GDL-309, Critical Mitigation Element Methodology Guideline

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- **Recommendations:** Shall be applied where design clarifications, further study, etc. are required by the evaluation team. Note that no credit shall be taken for 'Recommendations' when determining 'Residual Risk'; only 'Additional Controls' shall be considered in 'Residual Risk' determination.

Additional Controls and Recommendations shall be written to be standalone, e.g. understandable without reference to the related PHR worksheets. They shall be clear, concise and consistent.

The Risk Level of an Event Scenario is derived from the assigned consequence and Probability Levels which determine the position in the QatarEnergy-IS HSE Risk Matrix of each potential HSE consequence in the Event Scenario. HSE Risks that have not been fully evaluated and verified, but have been identified by experienced risk personnel, are **Potential Risks** and require the same response as an HSE Risk.

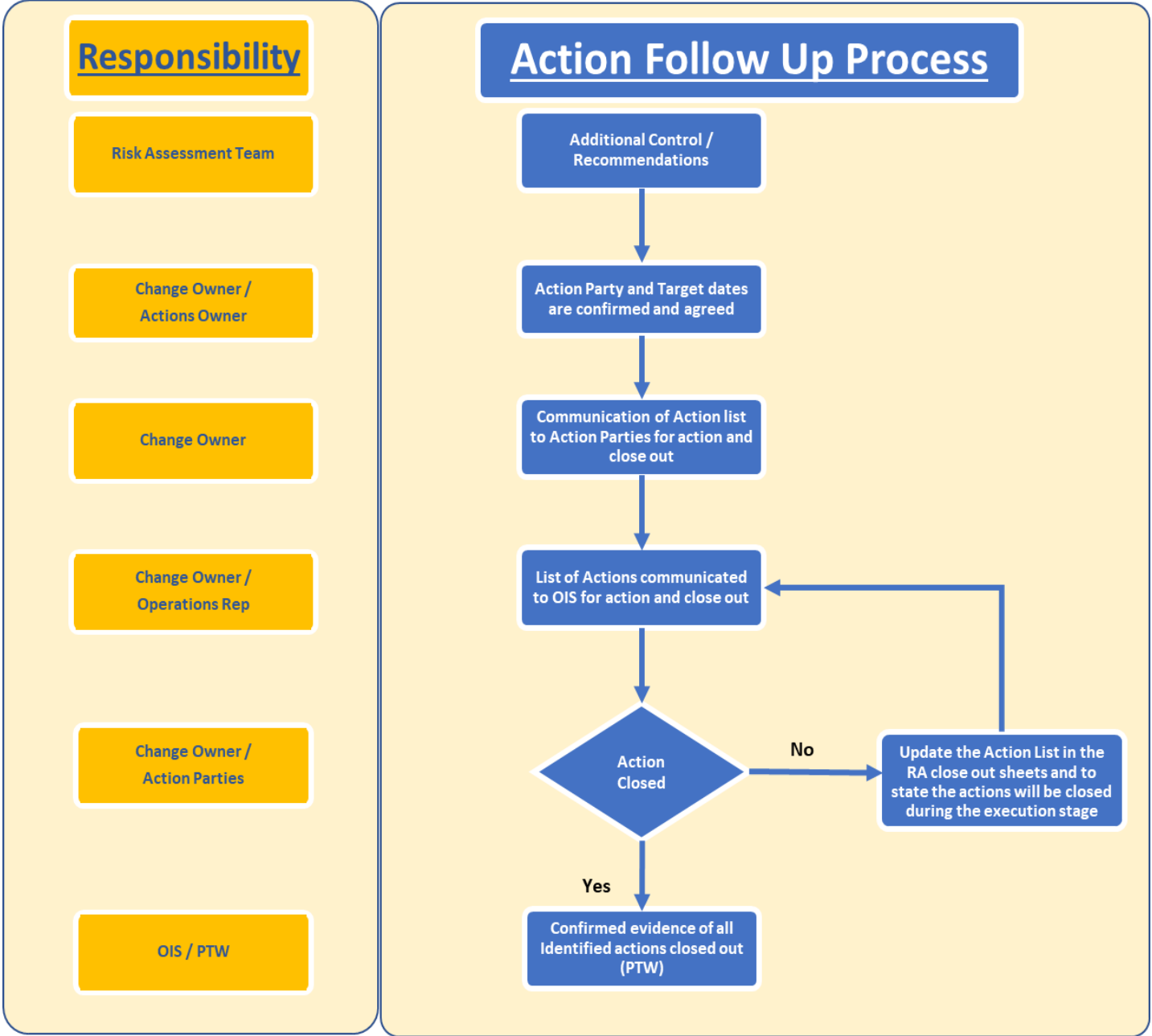
4.1.6 Action Items

Implementation of Action items - Additional Controls (CMEs and KMEs), and Recommendations, which result from PHRs (e.g. HAZOP, SIL, QRA etc.), shall be the responsibility of the identified actionee. Closure of the actions will be followed up using designated Action Tracker (i.e. QatarEnergy eHSE Action Tracker) until completion.

For What-if risk assessments: Risk assessment proponent to ensure the additional mitigations and recommendations coming out of the risk assessments are closed prior to the risk assessment subject/task being undertaken in the field. If the additional mitigations and recommendations cannot be closed prior to performing the task being undertaken in the field, proponent to provide a comment stating the status of the additional mitigations and recommendations (e.g. request order being raised, engineering work/construction work being raised, etc.). The action follow up process flowchart shall be followed in Figure 2

For all design modifications, the PSSR process requires, among other things, the issue of a 'RM Approvals Certificate', which confirms, in a stage-wise manner, that all relevant PHR's and resultant actions have been completed prior to the start-up of any project/system.

FIGURE 2. QATARENERGY-IS ACTION FOLLOW UP PROCESS



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4.1.7 Risk Matrix

The Risk Matrix shall be the method of approach for Risk Assessment in QatarEnergy-IS to address Consequence, Probability, Initial and Residual Risk levels for an Event Scenario and required response.

4.1.8 Potential Elevated Consequences (PEC)

Some Event Scenarios have much higher consequences than the maximum threshold value, i.e. a level 5 consequence, in the Risk Matrix. These very high consequences are referred to as Potential Elevated Consequences (PEC).

Each Event Scenario that is evaluated as having a Level 5 consequence shall also be evaluated for Potential Elevated Consequences and assigned one or more PEC codes based on the table below:

PEC 1	At this Event Scenario's assigned Probability, the potential consequence exceeds 14 Fatalities.
PEC 2	At this Event Scenario's assigned Probability, the potential consequence exceeds \$1 billion in combined Property Damage / Loss of Revenue.
PEC NA	PEC does not apply to this Event Scenario.

4.2 OPERATIONAL RISKS

HSE Risks are managed in QatarEnergy-IS through the QatarEnergy-IS HSE Management System (Health, Safety, and Environment Management System), and this is supplemented by Operations, Maintenance and Mechanical Integrity programs and systems.

As all HSE Risks cannot be anticipated, QatarEnergy-IS take the following additional measures to manage Operational Risks:

- In Safety briefings, Managers, Supervisors, Employees and Contractors shall be frequently reminded of their responsibilities to identify HSE Risks for which practicable Mitigation measures have not been implemented.
- Assign potential and identified Hazards to appropriate persons to evaluate, identify Mitigation options and implement appropriate Mitigation measures.
- All potential 'C with PEC', 'D' and 'E' Level Risks, have assigned consequences and probabilities and the Risk Matrix shall be used to determine the Risk Levels and appropriate response.
- Ensure that the QatarEnergy-IS MOC / PSSR programs are utilized.
- Marine Activities – A key element of QatarEnergy-IS operations relates to marine activities and associated risks. The hazard scenarios can be generically identified as follows:
 - a. Singular vessel incident (e.g. Fire, capsized).
 - b. Vessel to vessel collision.
 - c. Vessel to platform collision.
 - d. Personnel transfer (vessel to vessel, vessel to platform).
 - e. Punch through risk when vessels are jacking up.

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4.3 HSE RISK RESPONSE

Following a PHR, the PHR report is initially issued at revision 'A' for the PHR Team and RE Team review. Following receipt of comments and addressing of any comments the PHR report is then issued at revision 'B'. Should there be any 'C-with/PEC', 'D' or 'E' Risk Levels identified verification is required and following this the PHR report is then issued at revision 'C'.

The flowchart in Appendix C describes the What-if report process following a What-if risk assessment meeting. When a Potential or Verified and Validated 'C-with/PEC', 'D' or 'E' level risk is identified, immediate action must be taken to ensure that all practical measures are taken to assure existing mitigation is robust (e.g., operating parameters, AI/MI routines and set points/calibrations are verified, appropriate and up to date; interim operating options that would avoid or mitigate the risk are evaluated; etc.).

Risk Responses shall include one or more of the following actions, as applicable:

- Evaluation.
- Verification.
- Risk Validation.
- Communication to Management.
- Risk Register and Risk Inventory update.
- Action Tracking.
- Risk Re-evaluation and Reporting.

4.3.1 Verification⁴

When initial C level risk is identified in a PHR, verification of the Risk Level shall be carried out by the PHR team during the review itself, and duly noted, to ensure that all potential Mitigation measures have been considered and that Consequence and Probability Levels have been correctly assigned.

If a 'C-with/PEC', 'D' or 'E' Level HSE Risk is identified, a Risk Verifier from the Technical HSE Team shall evaluate the risk to determine if:

- The Risk Level and any PEC codes have been properly developed.
- More analysis is needed to confirm assumptions made to evaluate the risk.

This is required to ensure that correct Risk Evaluation of Event Scenarios has taken place and it also serves as a quality check of the evaluation, The Risk Verifier should also determine if any further mitigation can be applied.

Verifications carried out shall be documented, using the Risk Verification Form, and dependent on the nature and complexity, via an email or equivalent record, and filed electronically.

The Risk Verification shall be referenced to the MOC number, or equivalent ID, for the subject.

⁴ See IS-HSE-GDL-312, Risk Verification Guideline

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Following the verification process, any changes in register risk level shall require the register to be updated accordingly.

4.3.2 QatarEnergy-IS Risk Validation

All newly identified and / or changed 'C-with/PEC', 'D' and 'E' Level risks shall be entered and / or updated into HSE Risk registers by QatarEnergy-IS Technical HSE Team within 60 days after completion of the Process Hazard Review team group meetings.

4.3.3 QatarEnergy-IS Management Awareness / Acknowledgement

All verified 'C-with/PEC', and 'D' HSE Risk Levels shall, as soon as identified, be raised up to the QatarEnergy-IS, Field Operations Manager (OIF), HSE Manager (OIH), and Asset Manager for review and acknowledgement. Acknowledgement and approval shall be carried out by the QatarEnergy-IS Asset Manager (OI) for 'C-with/PEC', and 'D' HSE Risk Levels.

All verified 'E' HSE Risks shall be raised to the QatarEnergy-IS, Asset Manager (OI), Field Operations Manager (OIF) and HSE Manager (OIH) for review and acknowledgement. Acknowledgement and approval shall be carried out by the QatarEnergy Executive Vice President, Operations (VO) for 'E' HSE Risk Levels.

All verified 'C-with/PEC', 'D' and 'E' HSE Risk Levels shall undergo Risk Validation by the QatarEnergy-IS Technical HSE Team.

4.3.4 Communication

Risk Levels 'C-with/PEC', and 'D' must incorporate concurrence and approval from the QatarEnergy-IS Asset Manager (OI) who is required in order for the affected operation to continue (i.e., determine if risk is acceptable or unacceptable), if the HSE Risk cannot be reduced to a lower level within 60 days of verification. This concurrence must be renewed at least annually unless the HSE Risk is reduced to a lower level or the affected operation is discontinued.

For the highest Risk Level, 'E', approval of the QatarEnergy Executive Vice President, Operations (VO) is required to continue the affected operation (i.e., determine if risk is acceptable or unacceptable), if the HSE Risk cannot be reduced to a lower level within 60 days of verification. This concurrence must be renewed at least annually unless the HSE Risk is reduced to a lower level or the affected operation is discontinued, similar to Risk Levels 'C-with/PEC', 'D' above.

If the HSE Risk Level cannot be reduced to a lower level within 60 days of identification and verification, the HSE Risk shall be recorded in the QatarEnergy-IS HSE Risk Register, including identified hazards, risk evaluations, mitigation measures (KMEs and CMEs), and implementation status, reasons for actions taken or not taken, target Risk Level, and evidence of risk verification.

4.3.5 HSE Risk Register

QatarEnergy-IS shall record a new or updated HSE Risk Register, for all newly identified and/or changed 'C-with/PEC', and 'D' Level HSE Risks, and submit to QatarEnergy-IS Field Operations Manager (OIF), and HSE Manager (OIH) for review and acknowledgement and QatarEnergy-IS Asset Manager (OI) for acknowledgement and approval.

An HSE Risk Register shall be completed and submitted to the above for each Event Scenario with a verified C-with/PEC, D or E Risk Level that is not mitigated to a Risk Level below 'C-with/PEC' within the 60-day period allowed.

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Potential Risks must be processed and reported in the same manner as validated risks in order to assure that evaluation, mitigation (if appropriate) and communication are a management priority. The QatarEnergy-IS HSE Risk Registers are located in the HSE Risk Management Database.

HSE Risk Registers shall be reported to QatarEnergy. Corporate HSE BC (keep that) after translating the risk as per QatarEnergy Corporate Risk Matrix and shall be reviewed monthly during the HSE monthly management meeting review.

QatarEnergy-IS shall ensure that the HSE Risk Registers are kept current by revising them when HSE Risks are impacted by changes resulting from MOC programs, QatarEnergy-IS or industry incidents, and substantial fluctuations in the value of or revenue from the facility/system, new mitigation measures or any other circumstance that would be expected to affect the Risk Level.

QatarEnergy-IS shall create a new or updated HSE Risk Register, for newly identified and/or changed 'C-with/PEC', 'D' and 'E' Level HSE Risks. The HSE Risk Register shall be entered into the HSE Risk Register Database within 60 days of verification or a change in position on the QatarEnergy-IS HSE Risk Matrix. For risks identified in existing facility PHRs, the 60-day reporting time will start when the PHR team completes their group meetings, and review / verification, i.e. issue of revision 'C' (after verification).

For risks identified in MOCs or PHRs related to facilities that are not in service, the reporting time will be initiated when the facility is placed in service. In all cases, QatarEnergy-IS shall be consulted when there are atypical situations that justify extending the reporting deadline.

4.3.6 HSE Risk Register Action Tracking

Mitigation plans and target dates shall be entered in the Recommended Additional Safeguards/Control Measures column of the Risk Register.

Actions shall be admitted into an action tracking system and will have an ongoing review as actions are updated and closed. Requirements for entry into the QatarEnergy-IS tracking system shall apply for actions/recommendations related to recommendations to address 'C-with/PEC', 'D', and 'E' level risks.

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5.0 REFERENCES TO OTHER DOCUMENTS

Document Number	Title
IS-HSE-PRC-308	Process Hazard Reviews (PHR) Procedure
IS-HSE-GDL-303	RM Administration Guideline
IS-HSE-GDL-309	Critical Mitigation Element Methodology Guideline
IS-HSE-GDL-310	Developing Process Event Scenarios Guideline
IS-HSE-GDL-312	Risk Verification Guideline

TABLE 1: REFERENCES TO OTHER DOCUMENTS

6.0 DEFINITIONS

NIL

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APPENDIX

APPENDIX A - REVISION HISTORY LOG

TABLE 2: REVISION HISTORY LOG

Revision Number: 01

Date: 05/09/2023

Item Revised	Revision Description	Page No.
All	The document is updated based on the new format for Management system documents	
Remarks:		

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APPENDIX B - LIST OF ABBREVIATIONS

NIL

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APPENDIX C - IDD EL SHARGI RISK MATRIX

PROBABILITY	Quantitative Approach (frequency per year)	Industry History Approach	Critical Mitigation Element (CME) Approach	HSE Risk Matrix						
	Greater than 1x10 ⁻²	Unwanted event <u>has occurred</u> in this specific operation within this industry using this same technology	No CMEs to prevent an undesired Event Scenario	PROBABILITY	5	B	C	D	E	E
	Between 1x10 ⁻³ and 1x10 ⁻²	Unwanted event <u>almost certain</u> to occur in this specific type of operation within this industry using this same technology during the life of the operation	Single CME to prevent an undesired Event Scenario		4	B	C	D	D	E
	Between 1x10 ⁻⁴ and 1x10 ⁻³	Unwanted event <u>likely</u> to occur in this specific type of operation within this industry using the same technology during the life of the operation	2 CMEs, failure of 1 would not allow an undesired Event Scenario		3	A	B	C	D	D
	Between 1x10 ⁻⁵ and 1x10 ⁻⁴	Unwanted event <u>is unlikely</u> to occur in this specific type of operation within this industry using this same technology during the life of the operation	3 CMEs, failure of 2 would not allow an undesired Event Scenario		2	A	B	B	C	D
	Less than 1x10 ⁻⁵	Unwanted event is <u>Highly Unlikely</u> to occur in this specific type of operation within this industry using this same technology	4 or more CMEs, failure of 3 would not allow an undesired Event Scenario		1	A	A	B	B	C
* Risk Level = 'C' if <u>no</u> Fatality					1	2	3	4	5	
				CONSEQUENCE						
CONSEQUENCE	Personnel Human Injury <u>at</u> a facility		Minor injuries/illnesses (e.g. first aid)	Medical treatment, hospital admission, lost work days	1 Fatality	2 or 3 Fatalities	More than 3 Fatalities			
	Environmental Impairment A. Cleanup/Remediation cost B. Habitat or wildlife population		A. Less than \$1 M B. Limited, localized, temporary, and not severe (Less than \$500 K)	A. \$1 M to \$5 M B. Localized, short-term with no lasting consequences, slightly severe, and partial restoration / mitigation possible (\$500 K to \$2 M)	A. \$5 M to \$10 M B. Moderate, temporary, moderately severe, and partial restoration / mitigation possible (\$2 M to \$5 M)	A. \$10M- \$50M B. Major, long-term restoration / mitigation possible, and severe (\$5 M to \$25 M)	A. More than \$50 M or not possible B. Extensive, substantial, severe, and restoration not possible, permanent impairment (more than \$25 M)			
	Property and Revenue Facility property damage and loss of revenue (excluding cleanup/remediation) *		Less than \$1 M	\$1 M to \$20 M	\$20 M to \$50 M	\$50 M to \$250 M	> \$250 M			
	Impacts outside a facility A. Human Injury outside a facility (except security or transportation scenarios) B. Human Injury outside a facility related to security or transportation scenarios C. Damage to property not owned or operated by IS or a contractor (excluding cleanup/remediation)		A. Nuisance Impacts (e.g. smoke, noise, odor) B. Minor injuries/ illnesses (e.g. first-aid), no lost work days C. Less than \$100K	A. Minor injuries / illnesses (e.g. first-aid) B. Medical treatment, hospital admission, lost work days C. \$100 K to \$1 M	A. Medical treatment requiring emergency room visit OR shorter-term hospital admission B. 1 Fatality C. \$1 M to \$5 M	A. 1 longer-term hospital admission B. 2 or 3 Fatalities C. \$5 M to \$20 M	A. 1 Fatality B. More than 3 Fatalities C. More than \$20 M			

* For the purposes of applying the Risk Matrix to QP/IS plants and facilities, potential economic effects of an Event Scenario shall be estimated on a gross basis to reflect the Asset as a whole, including revenue & costs, as well as the property of Contractors. Use current year Asset Plan Gross US\$ Value for oil price.

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Risk Level	Risk Response	Reviews / Acknowledges (Initial / Annual)	Acknowledges / Signs (Initial / Annual)
A	No further action or risk evaluations required. Asset/New Venture shall document identified hazards, risk evaluations, and mitigation measures (KMEs and CMEs).	► IS - OIS	
B	IS shall : - Consider feasibility for practical and cost-effective additional risk evaluations and mitigation measures (KMEs and CMEs). - Document identified hazards, risk evaluations, and mitigation measures (KMEs and CMEs).	► IS - OIS	
C	IS shall : - Consider feasibility for additional risk evaluations and mitigation measures (KMEs and CMEs). - Verify Risk Level. - Document identified hazards, risk evaluations, mitigation measures (KMEs and CMEs), reasons for actions taken or not taken, and evidence of risk verification.	► Head of Site Operations (OIF/1) ► HSE Manager (OIH)	Field Operations Manager (OIF)
C with PEC & D	IS shall : - Perform appropriate qualitative or, if warranted, quantitative risk analysis to verify the level and identify additional mitigation measures (KMEs and CMEs). - Develop action plans and target dates for implementation of mitigation measures (KMEs and CMEs) considered to be feasible and determine the target Risk Level. - Document identified hazards, risk evaluations, mitigation measures (KMEs and CMEs), reasons for actions taken or not taken, and evidence of risk verification.	► Field Operations Manager (OIF) ► HSE Manager (OIH)	IS Asset Manager (OI)
E	In addition to the response elements required for a D Level Risk, concurrence from the Executive VP, Operations (VO) is required to continue the affected operation (i.e., determine if risk is acceptable or unacceptable).	► Field Operations Manager (OIF) ► HSE Manager (OIH) ► IS Asset Manager (OI)	Executive VP, Operations (VO)
PEC Level	Note: For 'Personnel' and 'Property & Revenue' categories with a Level 5 Consequence, one of the following PEC Codes shall be assigned. The Credible Scenario should not be modified.		
PEC 1	At this Event Scenario's assigned Probability, the potential consequence exceeds 14 Fatalities.		
PEC 2	At this Event Scenario's assigned Probability, the potential consequence exceeds \$1 billion in combined PD/LOR.		
PEC N/A	PEC does not apply to this Event Scenario.		
Notes:	1) In Case of C with PEC, D and E, if the HSE Risk cannot be reduced to a lower level within 60 days of Identification, IS shall translate "using agreed Translation matrix" the relevant Risks to QP Corporate Risk Levels and reported to QP Corporate HSE&Q if required. 2) Consequence should always be addressed first, before Probability; this precludes ruling out potentially high-consequence events because of perceived low probability. Consider the worst-case, reasonable, potential consequence for each category (e.g. Personnel, Environmental Impairment etc.) 3) CME approach is the preferred methodology for assigning probability levels for loss of containment Event Scenarios. CMEs must: be effective in preventing or mitigating the Event Scenario; be independent of other CMEs and not be compromised by the failure of other CMEs; have acceptable reliability, and meet proper equipment classification. 4) Industry History Approach should only be used when the CME approach is not appropriate. 5) No level of risk is considered 'acceptable' to IS; relatively low-level risks should be further reduced where simple, cost-effective controls can be added; i.e. the 'As Low As Reasonably Practicable' (ALARP) principle shall apply.		

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APPENDIX D – WHAT-IF REPORT PROCESS

